3	**(a)**	A body continues at rest or constant velocity unless acted on by a resultant (external) force	B1	[1]
	(b) (i)	constant velocity/zero acceleration and therefore no resultant force	M1	
		no resultant force (and no resultant torque) hence in equilibrium	A1	[2]
	(ii)	<u>component of weight</u> = 450 × 9.81 × sin 12° (= *917.8*)	C1	
		tension = 650 + 450 g sin12° = (650 + 917.8)	C1	
		= 1600 (1570) N	A1	[3]

(iii)	work done against frictional force or friction between log and slope	M1	
	output power greater than the gain in PE / s	A1	[2]

4 **(a)** total resistance = 30 (kΩ) C1